

Real-world walking economy: can laboratory equations predict field energy expenditure?

Summary

This study investigated the accuracy of laboratory-based equations in predicting field walking energy expenditure. Seven subjects walked a 6,415-meter field course with varying gradients and terrain, while wearing portable calorimeters and GPS units. The study found that while laboratory equations can predict field walking energy expenditure, the accuracy is equation-dependent and can vary significantly.

Abstract

We addressed a practical question that remains largely unanswered after more than a century of active investigation: can equations developed in the laboratory accurately predict the energy expended under free-walking conditions in the field? Seven subjects walked a field course of 6,415 m that varied in gradient (-3.0 to $+5.0\%$) and terrain (asphalt, grass) under unloaded (body weight only, W_b) and balanced, torso-loaded ($1.30 \times W_b$) conditions at self-selected speeds while wearing portable calorimeter and GPS units. Portable calorimeter measures were corrected for a consistent measurement-range offset ($+13.8 \pm 1.8\%$, means $\pm$ SD) versus a well-validated laboratory system (Parvomedics TrueOne). Predicted energy expenditure totals (mL O_2 /kg) from four literature equations: ACSM, Looney, Minimum Mechanics, and Pandolf, were generated using the speeds and gradients measured throughout each trial in conjunction with empirically determined terrain/treadmill factors (asphalt = 1.0, grass = 1.08). The mean

energy expenditure total measured for the unloaded field trials (981 ± 91 mL O_2 /kg) was overpredicted by +4%, +13%, +17%, and +20% by the Minimum Mechanics, ACSM, Pandolf, and Looney equations, respectively (corresponding predicted totals: $1,018 \pm 19$, $1,108 \pm 26$, $1,145 \pm 37$, and $1,176 \pm 24$ mL O_2 /kg). The measured loaded-trial total ($1,310 \pm 153$ mL O_2 /kg) was slightly underpredicted by the Minimum Mechanics equation (−2%, $1,289 \pm 22$ mL O_2 /kg) and overpredicted by the Pandolf equation (+13%, $1,463 \pm 32$ mL O_2 /kg). Computational comparisons for hypothetical trials at different constant speeds (range: 0.6–1.8 m/s) on variable-gradient loop courses revealed between-equation prediction differences from 0% to 37%. We conclude that treadmill-based predictions of free-walking field energy expenditure are equation-dependent but can be highly accurate with rigorous implementation.

NEW & NOTEWORTHY Here, we investigated the accuracy with which four laboratory-based equations can predict field-walking energy expenditure at freely selected speeds across varying gradients and terrain. Empirical tests involving 6,415-m trials under two load conditions indicated that predictions are significantly equation dependent but can be highly accurate (i.e., $\pm 4\%$). Computations inputting identical weight, speed, and gradient values for different theoretical constant-speed trials (0.6–1.8 m/s) identified between-equation prediction differences as large as 37%.

Keywords: biomechanics, generalized equation, locomotion, terrain, wearable sensors

INTRODUCTION

The energetic and mechanical demands of walking in natural settings are of seminal importance to human biology. The body's morphology, health, and physiological status are all fundamentally influenced by walking dynamics.

Consequently, considerable anthropological, physiological, biomechanical, and ecological attention has been devoted to human walking. However, the extensive investigative history of gait metabolic requirements has relied predominantly upon data from highly controlled conditions, most often treadmills within research laboratories. Considerably less attention has been devoted to the economy of human walking in the natural world.

Accurate understanding of human walking metabolism in field settings has broad scientific relevance. Natural historians consider the relationship between walking economy and environmental topography to be a primary determinant of the physical evolution of the human body (1–3). Practically, accurate predictive capabilities would serve applications that range from quantification of macronutrient needs for hikers to feasibility and mission readiness projections for foot soldiers and wildland firefighters. The latter two subpopulations are often required to transport loads and traverse variable grades and terrains to execute essential, urgent, and even life-and-death tasks (4). With the emergence of wearable and monitoring technologies and current availability of detailed contour maps to support field predictions, accurate field predictive algorithms are arguably of greater practical importance than ever.

Dozens of equations for predicting human walking metabolism exist. However, those incorporating the speed, grade, and load carriage capabilities needed for field-condition estimates are few. The most established is the equation by Pandolf et al. (5) which was originally formulated from several relatively small treadmill and overground walking datasets (6–8) to include speed, load, and positive-only gradient capabilities. The equation was appended by Santee et al. (9) to include negative gradients several decades later. The long-standing, frequently used equation of the American College of Sports Medicine (ACSM, 10) also evolved from a number of smaller datasets (11, 12) to include speed and

positive-only gradient capabilities. However, the ACSM equation does not presently incorporate either negative gradients or load carriage. A more recent equation was formulated by Looney et al. (13) from a sizeable, literature-aggregated set of laboratory treadmill data. The Looney equation encompasses a broad range of positive and negative gradients but does not incorporate load carriage. Finally, the recent Minimal Mechanics model (14) was developed from a single, relatively large, original treadmill database that included several load conditions, a broad range of speeds, and both positive and negative gradients.

From a practical standpoint, a noteworthy feature of these leading laboratory equations is that accurate field translation is broadly assumed (5, 13–15) despite the full absence of testing on humans walking freely in natural settings. Several differences between laboratory treadmill and free-walking field walking conditions could introduce predictive error. These include potential across-condition differences in balance requirements, speed variability, and terrain. In the last case, significant elevations in walking metabolism above treadmill values on loose and slippery terrain are well documented (6, 16), but asphalt-treadmill comparisons have been inconsistent (17, 18) and consideration of other relevant surfaces, like grass, have been limited. Accordingly, current incorporations of terrain effects into laboratory prediction equations (5) are also uncertain.

Presently, the walking metabolic data available from the field to evaluate the predictive accuracy of existing equations are extremely limited. The rigorous field work by de Mullenheim et al. (19), by design, involved many brief trials on selected grades under constant-speed and grade conditions. The progressive field efforts of Looney et al. (20, 21) involved longer trials undertaken by heavily loaded subjects instructed to walk as fast as possible. In both studies, the accuracy limitations of the portable calorimeters used to acquire metabolic data in the field were responsibly acknowledged by the

authors, but not quantified. The calorimeter model in question consistently exceeds well-validated laboratory systems under similar conditions by magnitudes ranging from 2% to 25% (22–28). Thus, experimental design differences and uncertainties limit the application of these studies to the specific question of field accuracy addressed here.

We undertook the present study of the energetics of both loaded and unloaded walking at self-selected speeds in the field to test two hypotheses. First, we hypothesized that the free-walking energy expenditure measured in the field could be accurately predicted using existing laboratory equations. This expectation was based on the limited across-condition data available suggesting that any differences between the energetic requirements of treadmill and overground walking on firm terrain are relatively small and quantifiable (18). Second, we hypothesized that the predictions of existing field-capable laboratory equations using identical speed and grade data from the field would not differ from one another. This expectation was based simply on the common reliance of these equations on laboratory treadmill walking metabolic data acquired with standardized indirect calorimetry techniques.

METHODS

Experimental Design

We tested our first hypothesis that laboratory treadmill prediction equations would provide accurate estimates of field walking energy expenditure using an experimental approach that incorporated the following strategies. First, we identified a field walking course that had level, positive and negative gradients, and different types of terrain. Second, we acquired metabolic data across course-relevant terrain conditions to directly determine possible treadmill versus overground offsets empirically. Third, we measured the

performance of the portable calorimeter used in the field versus a well-validated laboratory metabolic system under the same conditions in our laboratory. Fourth, we had subjects who undertook relatively long trials with simultaneous acquisition of oxygen uptake and position data throughout. Fifth, we administered both unloaded and loaded trials to incorporate conditions that would alter both the measured and predicted walking energy expenditure totals acquired.

We tested our second hypothesis that the predictions of existing laboratory equations would not differ from one another using both empirical and computational-only approaches. The empirical approach compared the energy expenditure totals predicted by the four literature equations from the same sets of speed, gradient, and weight (body + load) data acquired during the unloaded and loaded field trials. The computational-only approach compared the energy expenditure totals predicted by the four equations for hypothetical trials across a range of constant speeds on loop courses of differing gradient profiles. This computational approach allowed us to extend prediction equation comparisons well beyond the specific grades and walking speeds measured during the experimental field trials.

Laboratory Treadmill Equations from the Literature

We selected the four generalized treadmill prediction equations we deemed most relevant and useful for predicting field walking energy expenditure. Two of these equations, those of the ACSM and Pandolf et al. have been in the literature for decades and have been tested for accuracy by many investigators, largely under laboratory conditions. The other two equations, the Looney et al. and Minimum Mechanics or Ludlow/Weyand equations, are both more recent, but have been included because they incorporate the relatively broad speed and gradient conditions often encountered in the field. Each of the four equations was used to generate predictions of the

total energy expended during field trials that are here expressed in units of oxygen uptake (mL O₂/kg body wt).

Because the Pandolf/Santee and Looney et al. equations predict energy expenditure in energy units while the ACSM and Minimum Mechanics equations predict oxygen uptake rates, an energetic equivalent for oxygen was necessary for between-equation comparisons. Here, we report results in units of oxygen uptake throughout the manuscript per common physiological practice. The formulas for the four equations, including the necessary conversion factors, are detailed in Table 1. At present, the practical consequences of between-equation formulaic differences, or lack thereof in field settings, has not been well established.

Table 1.

Laboratory treadmill predictive equation formulas expressed in oxygen units

Algorithm	Grades	Equation— $\dot{V}O_2$ (mL/kg/min)
Minimum Mechanics	≥ 0	$3.05 + (w+l)wt(0.32g + 3.28 + (1 + 0.19g)2.66s^2)$
	< 0	$3.05 + 0.73(w+l)wt(3.28 + 2.66s^2)$
Looney		$2.985(1.44 + t(1.94s + 0.43 + 0.24s^4 + 0.34sg(1 - 1.051 - 1.1g + 32)))$
Pandolf	≥ 0	$2.985w(1.5w + 2(w+l)(lw)^2 + t(w+l)(1.5s^2 + 0.35sg))$
	< 0	$2.985w(1.5w + 2(w+l)(lw)^2 + t(w+l)(1.5s^2) + t(g(w+l)s3.5 - (w+l)(g+6)2w + (25-s^2)))$
ACSM	≥ 0	$t(0.1\sigma + 1.8\sigma\gamma) + 3.5$
	< 0	$t0.1\sigma + 3.5$

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s = speed (m/s), g = grade (%)—(0 to 100), t = terrain factor, w = body

weight (kg), l = load (kg), σ = speed (m/min), γ = grade (%)—(0 to 1).

Conversions from energy to O_2 units were implemented for the Pandolf and Looney equations using a conversion factor of 20.1 mL of O_2 per Joule per Blaxter (31). ACSM, American College of Sports Medicine.

Course Characteristics

The field course was located in and near Flag Pole Hill Park in Dallas, TX. Trials were initiated on a level concrete access trail leading into the park; once inside the park, subjects completed four similar loops that included ascending and descending sections of Flag Pole Hill. The ascending portions occurred on asphalt on the first and third loop, and on grass on the second and fourth loops. Descending and level portions followed similar grassy paths on all loops with some individual variation occurring in how subjects chose to navigate between course landmarks. The total distance was just under 6,400 m, or 4.0 mi, on average with the majority occurring on well-maintained grass. The course elevations and gradients were established from the GPS position data acquired throughout the walking trials and during standing trials at selected course locations. The terrain and elevation profile of the course is shown in Fig. 1.

Figure 1.

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Field course elevation and terrain versus distance profiles. Surface terrains are illustrated as gray for pavement,

brown for packed dirt, black for asphalt, and green for grass. Elevations are referenced to sea level. Gradient labels represent % grades. The inset provides the vertical and horizontal distances corresponding to 25 meters on the the respective axes.

Subjects and Testing

All subjects provided written, informed consent in accordance with the protocol approved by the Institutional Review Board of Southern Methodist University. Due to the volume of field and laboratory testing required, separate subject groups were recruited for: 1) terrain testing ($n = 9$), 2) Douglas bag versus laboratory metabolic system testing ($n = 15$), and 3) field walking trials at the Flag Pole Hill course ($n = 7$).

Terrain observations were acquired from a group of nine subjects who completed level walking trials on the treadmill, asphalt, and grass surfaces. These subjects were healthy, active adults between 18 and 40 yr of age ($n = 6$ females, $n = 3$ males, 74.6 ± 14.8 kg). Evaluation of the accuracy of the laboratory metabolic system (Parvomedics TrueOne) occurred via comparison with the gold standard Douglas bag method under identical laboratory treadmill walking conditions. For this purpose, data were acquired from 15 active adults between 18 and 40 yr of age ($n = 9$ females, $n = 6$ males, 74.5 ± 15.0 kg). Direct validation of the laboratory metabolic system was necessary for full confidence in the interpretation of the offset between the field and laboratory calorimeters.

The field-testing subject group of seven subjects completed field-walking trials in two load conditions and accompanying laboratory sessions that required valid field assessments. This field-testing group subjects were between 22 and 40 yr of age ($n = 1$ female, $n = 6$ males). Their physical characteristics, as well as their resting and maximal rates of aerobic metabolism are detailed in Table 2. Each of the field test subjects completed a minimum of five test sessions as follows: 1) treadmill determination of

maximal aerobic power ($\dot{V}O_{2\max}$) using a progressive-grade protocol at a running speed selected to elicit volitional failure in 10 min or less, 2) steady-state level treadmill walking trials across a wide range of speeds, 3) an unloaded field walking trial, 4) a loaded field walking trial, and 5) laboratory measurement of resting and standing metabolism. The nonexercise metabolic measurements were acquired in the early morning in a fasted state. Subjects were instructed to fast for at least 8 h, refrain from caffeine ingestion and to move no more than necessary before reporting to the laboratory for testing. After arriving, subjects completed 30 min of supine resting followed by 15 min of quiet standing.

Table 2.

Field subject physical characteristics and metabolic measures

Subject No.	Age, yr	Height, cm	Weight, kg	RMR, mL O ₂ /kg·min	$\dot{V}O_{2\max}$, mL O ₂ /kg/min
29	20	185	75.9	3.59	44.8
47	40	165	100.3	3.12	40.1
64	22	176	81.6	3.59	39.0
74	35	164	62.0	3.09	42.9
75	34	180	103.5	3.32	46.7
77	22	173	75.5	3.78	41.4
78	32	181	96.5	3.17	52.2
Means ± SD	29.3 ± 7.8	174.9 ± 8.0	85.0 ± 15.4	3.38 ± 0.3	43.9 ± 4.6

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In one of the seven loaded field trials, the GPS system failed to acquire position data over the last third of the trial. Because the subject was not available for retesting, loaded condition data were available for only six

subjects.

Terrain versus Treadmill Energetics

Walking energy expenditure was measured for the nine terrain subjects at speeds of 1.0, 1.3, and 1.6 m/s on grass and asphalt overground surfaces as well as on the laboratory treadmill (Trackmaster TMX22, Full Vision Inc., Newton, KS). We used a well-validated laboratory metabolic system (ParvoMedics TrueOne, Sandy, UT) that relies on a pneumotachometer to measure volume flow rates and paramagnetic and infrared sensors to analyze fractional concentrations of O_2 and CO_2 , respectively. Gas analyzers were calibrated using room air and calibration gas of known O_2 and CO_2 concentrations. Douglas bag volumes were measured with a dry gas meter (Harvard Apparatus, Holliston, MA) with simultaneous temperature reading after O_2 and CO_2 concentration determination. Gas volumes were converted to STPD values.

All walking trials were a minimum of 5 min in duration. Steady-state rates were determined by averaging the single-minute rates of oxygen uptake for each of the last two minutes. Trials at the three speeds were performed twice under each condition to acquire duplicate measures of trial oxygen uptake. The duplicate measures at each speed were averaged to determine the final values for each subject for the treadmill and overground conditions.

Overground trials took place on level 50-m oval courses setup in an asphalt parking lot and in a well-maintained field with short grass, respectively. Metronome sounds at intervals of 30 s or less and verbal feedback were provided for maintenance of the desired overground trial speed. Subjects wore a custom lightweight backpack frame designed to support the Douglas bags on their backs that allowed easy valve access for opening and closing. Gas was collected for 60 s in each bag by having a trailing investigator open

and close the valve at the designated times.

Portable versus Laboratory Calorimeter

The oxygen uptake measures of the portable and laboratory calorimeter systems were compared during treadmill walking at identical speeds ranging from 0.4 to 1.6 m/s. Also compared were: 1) the standing rates of oxygen uptake acquired at the beginning of each field trial from the portable calorimeter versus those acquired in the laboratory, and 2) steady-state walking metabolic rates acquired from the portable calorimeter during the initial portion of the field course on a firm level surface versus laboratory treadmill values acquired at similar walking speeds.

Laboratory Calorimeter versus Douglas Bag Technique

Also evaluated was the agreement between the gold standard Douglas bag technique and the Parvomedics TrueOne system by having the same subjects ($n = 15$) walk at the same three treadmill speeds of 1.0, 1.3, and 1.6 m/s to acquire steady-state measurements under identical laboratory conditions.

Field Trials

For all field walking trials, metabolic data were acquired with the Cosmed K4b₂ (COSMED, Rome, Italy) portable calorimeter. Gas exchange data were acquired on a breath-by-breath basis throughout the test and averaged over 15-s intervals for analysis. Heart rate data were collected using a Polar RS400 (Polar Electro Oy, Kempele, Finland) multi-sport watch. The Kestrel 4500 Weather and Environmental Meter (Nielsen-Kellerman, Boothwyn, PA) was used to acquire barometric pressure, ambient temperature, and humidity. For subject safety and optimal performance of the portable

calorimeter, tests were not conducted unless the ambient temperature was between 12.8 and 24.4°C and the relative humidity was less than 90%.

Position data were acquired using a Trimble GPS system (Trimble Navigation Limited, Westminster, CO) consisting of a Trimble Pro 6H series receiver unit and Trimble Juno 3b series handheld unit. The total weight of the receiver (1.04 kg) and backpack (1.36 kg) was 2.40 kg. The unit was initialized during quiet standing that preceded each walking trial. A minimum of seven satellites were successfully accessed throughout all the field-walking trials. Position data were acquired at 1.0 Hz and stored in the standard storage format (SSF) onsite before subsequent differential correction using the Texas Department of Transportation (TxDOT), Dallas base station. The minimum accuracy of the corrected data is 0.1 m for the horizontal position, and 0.5 m for vertical position.

Upon arrival at the field site, subjects were weighed with a Detecto DR400 portable scale and fitted with the Cosmed K4b₂, Trimble GPS unit, a Polar watch, and a standardized shoe (Brooks Cascadia). The portable calorimeter was calibrated on-site before the beginning of each test. For loaded condition trials, loads corresponding to 30% of body weight were added using a vest equipped with pouches and a MOLLE style backpack for even anterior-posterior load distribution as previously described (14). Tests began with the subject standing for 5 min after the Cosmed K4b₂, Polar, and Trimble units were initialized. Walking began from a position on the level concrete access trail just over 800 m from the park perimeter with subjects proceeding into the park and along the course as previously described (Fig. 1). Subjects were instructed to walk at a self-selected comfortable pace throughout and to pause for resting or load-relief if they chose to. In practice, some individual subjects did elect to pause briefly at one or more junctures, predominantly during the loaded trials; none chose to dismount the load. Two trailing spotters monitored the subject and the GPS and

calorimeter data streams throughout each trial. Subjects were directed to the course landmarks as needed to enable them to complete the designated looped portions of the course. Trials were terminated when the subject finished the fourth course loop.

Field Energy Expenditure

Field trial energy expenditure was determined from oxygen uptake acquired throughout each trial by the portable calorimeter. Given the submaximal nature of the nonsteady-state exercise intensities involved, the oxygen uptake totals acquired over the roughly 70-min walking trials provide valid measures of the total energy expended (29, 30). Field respiratory exchange ratios ($\dot{V}CO_2/\dot{V}O_2$) were determined from the measured total volumes of CO_2 liberated and O_2 taken up over the course of each trial.

The predictive totals for the Pandolf/Santee and Looney et al. equations were converted from Watts, or Joules per second, to units of oxygen uptake using a conversion factor of 20.1 J per mL of O_2 (31). The intermediate energetic equivalent of oxygen value of 20.1 was selected in accordance with the substrate oxidation blend of fat and carbohydrate expected for lengthy, continuous exercise of low-to-moderate intensity. For reference, the 20.1 value corresponds to a substrate mixture yielding a nonprotein respiratory quotient ($\dot{V}CO_2/\dot{V}O_2$) slightly greater than 0.80.

Measured Field Oxygen Uptake versus Laboratory-Equation Predictions

The total oxygen taken up during the walking trials was measured using the portable calorimeter and corrected for the offset measured versus the Parvomedics TrueOne laboratory system established under laboratory treadmill conditions.

The oxygen uptake totals (mL O₂/kg) predicted from the four literature equations were determined using the total weight (body + load), GPS-measured speeds and gradients, and location-specific terrain factors (Fig. 1). The course positioning data provided by the GPS were verified by investigator time recordings of when subjects passed designated points along the course. Walking speeds were averaged over each 15-s interval of the field trials. Constant gradient inputs were used for the level (0% for the park access road, hilltop, and bottom segments of each loop), ascending (+4.6 asphalt or +4.9% grass), and descending (−3.0%) course sections, respectively.

The influence of terrain on the equation values generated from total load, speed, and grade was implemented in accordance with the course locations illustrated in Fig. 1. Empirical determinations reported subsequently in the results identified terrain/treadmill factors of 1.0 for asphalt and 1.08 for the grass. The mixed asphalt and hard-packed dirt course section between the concrete access trail and beginning of the first loop within the park covering roughly 600 m (gradient < 0.7%) was treated as asphalt with a gradient of zero for equation predictive purposes. In addition, the terrain factors determined here were only applied to the nonresting or nonstanding portions of the total energy expended as detailed in Table 1. This approach avoids the interpretation difficulties introduced when differing equation-specific terrain factors are applied to the same set of field circumstances (32).

Total energy expenditure predictions were made for all four literature equations for the unloaded field trials. Because the ACSM equation lacks downhill gradient capabilities, predictions for this equation necessarily required treating the downhill gradients as a gradient of zero. However, a modified ACSM-predicted total was determined by implementing the simple downhill gradient approach identified by Ludlow and Weyand (14) of multiplying the walking, or nonresting, portion of the body's total metabolic

rate under level conditions by 0.73 on downhill gradients. Total energy expenditure predictions and comparisons for the loaded trials were only made directly for the Minimum Mechanics and Pandolf/Santee equations because neither the ASCSM nor Looney et al. equations incorporate load. The more complex formulation of the Looney equation disallowed a straightforward modification of this nature to incorporate load.

Comparing Literature Equation Outputs Using the Same Speed and Gradient Inputs

Total oxygen uptake values were determined computationally for numerous, hypothetical, constant-speed field trials. Predictions were generated for three variable-gradient loop courses of 6,400 m or just under 4.0 miles: one with our measured course gradient (moderate-grade course, Fig. 1), one with the up- and downhill gradients on each loop of our original course halved (shallow-grade course), and one with our up- and downhill gradients on each loop of our original course doubled (steeper-grade course). For each of these three courses, predicted totals were determined for constant speed trials at increments of 0.1 m/s at speeds from 0.6 to 1.8 m/s. Respective equation outputs were determined from body weight, walking speed, and grade per the original formula of each equation provided in Table 1 and a terrain factor of 1.0 throughout. Percent differences between equations were calculated as the between-equations difference divided by the mean times 100, thus:

$$\{(\text{Eq. 1} - \text{Eq. 2}) / [(\text{Eq. 1} + \text{Eq. 2}) / 2]\} \times 100 \quad (1)$$

Statistical Analysis

Comparisons of the total oxygen uptake acquired from the portable calorimeter during the field walking trials versus the totals predicted by the

laboratory equations were assessed using a one-way ANOVA with repeated measures ($\alpha = 0.05$) with Bonferroni adjusted comparisons. For the unloaded field trials, measured total oxygen uptake values and those predicted by four equations: ACSM, Looney et al., Minimum Mechanics, and the Pandolf/Santee, were compared. For the loaded field trials, the measured means were compared only to the Minimum Mechanics and Santee/Pandolf equations because neither the ACSM nor Looney et al. equations incorporate load.

RESULTS

Terrain versus Treadmill Energetics

Walking rates of oxygen uptake for both overground conditions investigated, pavement and grass, had differing relationships to our treadmill-measured values (Fig. 2). The rates measured on grass, on average, exceeded those measured on the treadmill by a factor of 1.06, with larger differences being present at the faster two speeds. The oxygen uptake rates measured on pavement were closer in value to those measured on the treadmill, averaging 0.97 across the three speeds, and being nearly equal at the fastest two speeds.

Figure 2.

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Rates of oxygen uptake (means $\pm$ SD) measured using the Douglas bag method during level walking on pavement, short, well-maintained grass, and a laboratory treadmill at the same three speeds of 1.0, 1.3, and 1.6 m/s. Black and gray horizontal bars represent the range of individual, course-averaged walking speeds field subjects selected during their unloaded and loaded trials, respectively.

At the two faster, field-relevant speeds, the respective pavement/treadmill and grass/treadmill terrain factor ratios were 0.99 and 1.08. Given, the trial-to-trial variability, a final terrain-factor value of 0.99 was rounded up to 1.00 for simplicity before being implemented on the pavement portions of the field course while the numeric average at the two faster speeds of 1.08 was implemented for the grass portions of the course.

The relationship between oxygen uptake rates on pavement versus grass was relatively constant across the three speeds, with grass values being 1.09 times greater than pavement values.

Portable versus Laboratory Calorimeter

The rates of oxygen uptake measured by the portable versus laboratory calorimeter during laboratory treadmill walking at speeds from 0.6 to 1.6 m/s are shown in Fig. 3. Across the six test speeds administered, the values measured by the portable calorimeter were greater at every speed with a relatively constant percentage offset $[(\text{portable} - \text{laboratory}) / \text{laboratory}] \times 100$. At each speed, the offset fell within a range from 11.4% to 16.5% (means $\pm$ SD: $13.8 \pm 1.8\%$). Similarly, the percentage offset between the mean standing rate of oxygen uptake measured by the portable calorimeter at the 5-min stand at outset of the unloaded field trials and that measured by the laboratory calorimeter during the first 5-min of standing in the laboratory test was +14.3% (4.22 ± 0.47 vs. 3.69 ± 0.42 mL O₂/kg·min; see Y-intercept, Fig. 3).

Figure 3.

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Rates of oxygen uptake (means $\pm$ SD) during level treadmill walking measured by the portable calorimeter (open circles, dashed line) were $13.8 \pm 1.9\%$ greater than those measured by the laboratory calorimetry (closed circles, solid line) system at the same treadmill speeds. Oxygen uptake rates measured during quiet standing (open vs.

closed triangles) differed by a similar margin, being 12.5% greater for the portable vs. laboratory calorimeter. Unloaded and loaded rates of oxygen uptake [$\text{mL O}_2/\text{kg}$ total load (body + load), smaller and larger open triangle] during walking on dry, level concrete exceeded treadmill walking measures acquired from the laboratory calorimeter, but were similar to treadmill values acquired with the portable calorimeter.

The mean rates of oxygen uptake measured by the portable calorimeter during unloaded and loaded overground walking on the initial flat, concrete portion of the field course (smaller open triangles, Fig. 3) also exceeded those measured by the TrueOne laboratory system during treadmill walking at similar speeds. When expressed on a per kg total basis (body + load), the mean rate of oxygen uptake during both unloaded (1.35 m/s) and loaded (1.34 m/s) level overground walking was $13.0 \text{ mL O}_2/\text{kg}\cdot\text{min}$ for both load conditions.

The overground walking speed subjects selected on the initial flat portion of the course during the loaded trials (larger open triangles, Fig. 3) was nearly identical to the unloaded speed (1.34 vs. 1.35 m/s). At this similar speed, the mean rate of oxygen uptake measured by the portable calorimeter, when expressed on a per kg total load supported (body + load) basis for the loaded trials was $13.0 \text{ mL O}_2/\text{kg}\cdot\text{min}$, and therefore equal to the unloaded overground per kg value (open triangles vs. dashed line, Fig. 3).

Laboratory Calorimeter versus Douglas Bag Technique

Mean walking rates of oxygen uptake at each of the three test speeds were nearly identical when measured with the Parvomedics TrueOne laboratory metabolic system versus the Douglas bag method (Fig. 4). The overall mean values acquired with two techniques for the 15 participating subjects differed by $\leq 0.3 \text{ mL O}_2/\text{kg}\cdot\text{min}$ at each of the three protocol speeds. The disagreement between methods, when considered on an individual subject basis, only exceeded $1.0 \text{ mL O}_2/\text{kg}\cdot\text{min}$ in six of the 45 total observations made (15 subjects $\times$ 3 speeds). The mean absolute differences observed

between measurement methods for the 15 individual subjects at each of the respective speeds ranged from: 0.32 ± 0.44 to 0.87 ± 0.53 mL O₂/kg·min, respectively.

Figure 4.

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Measured rates of oxygen uptake acquired using both the Douglas bag method and the laboratory calorimeter at the three treadmill walking speeds: 1.0, 1.3, and 1.6 m/s. The values measured using the two systems differed by an average of $0.4 \pm 0.2\%$ at the three measurement speeds.

Field Trial Results

The mean walking speeds subjects selected during the unloaded and loaded walking trials were 1.40 ± 0.11 m/s (range: 1.20–1.57 m/s) and 1.32 ± 0.11 m/s (range: 1.18–1.47 m/s), respectively. In both cases there was relatively little within-trial variability in speed as subjects opted to maintain relatively similar speeds on the flat, inclined and declined portions of the field course. Moderate variation in the paths subjects chose between the course

landmarks to which they were directed resulted in an overall means $\pm$ SD course walking distance of $6,415 \pm 129$ m.

Representative oxygen uptake rate measures from the portable calorimeter during field trials for one subject in both the unloaded (continuous black line) and one loaded condition (continuous gray line) are shown in Fig. 5. Also illustrated are the corresponding rates of oxygen uptake per kg body mass predicted for these trials by one laboratory prediction equation, the Minimum Mechanics equation, for the unloaded (dashed black curve) and loaded conditions (dashed gray line) on the basis of the measured walking speed, gradient, and total weight supported (body + load). For both the unloaded and loaded trials, the measured and predicted rates of oxygen uptake per kg body mass were greater on the inclined portions of the course and lower on the declined portions in comparison to the level sections at the outset of the course and within each of the four loops. For the six subjects completing both load conditions, the total oxygen taken up for the loaded trials exceeded those for the unloaded trial condition by a factor of 1.35 ($1,310 \pm 153$ vs. 972 ± 95 mL O₂/kg).

Figure 5.

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Measured rates of oxygen uptake as a function of walking time during unloaded (gray solid line) and loaded (black solid line) walking trials in a single subject. Predicted rates of oxygen uptake (dashed lines, Minimum Mechanics equation) generated from the walking speed and grade data acquired from the GPS system during each of the respective trials are provided for reference. Measured rates of oxygen uptake were corrected as detailed (Fig. 3 and methods) and are expressed in relation to body mass in kg on a per minute basis.

The measured mean respiratory exchange ratio ($\dot{V}\text{CO}_2/\dot{V}\text{O}_2$) from the field trials was 0.82 ± 0.03 for the unloaded trials, and 0.83 ± 0.04 for the loaded trials. The overall mean from both load conditions of 0.83 ± 0.04 corresponds to a non-protein energetic equivalent of 20.2 Joules per mL of O_2 that closely matched the 20.1 value selected a priori based on the expected sources of substrate oxidation for walking. These field data indicate the maximum potential error introduced by conversion between the energy units in the Pandolf/Santee and Looney equations into oxygen units in the ACSM and Minimum Mechanics equations was less than 1.0%.

Group mean total oxygen uptake values ($981 \pm 91 \text{ mL O}_2/\text{kg}$) obtained for the seven subjects from their unloaded trials are shown in Fig. 6 (black dashed

horizontal line, *left* portion of the panel). The agreement between the mean field value obtained from the portable calorimeter for the field trials and the totals predicted by the laboratory equations varied [ANOVA: $F = 3,794.6$, $df = 4$]. The values predicted from the Minimum Mechanics equation were 4% greater than those measured ($1,018 \pm 19$ mL O₂/kg); those predicted with the ACSM ($1,108 \pm 26$ mL O₂/kg) and Pandolf/Santee ($1,145 \pm 37$) equations were 13% and 17% greater than the measured mean, respectively, and those from the Looney et al. equation prediction were 20% greater ($1,176 \pm 24$ mL O₂/kg). Differences between measured field values and equation-predicted totals were not statistically different for the Minimum Mechanics and ACSM equations ($P > 0.99$ and $P = 0.158$, respectively) but were significantly different for the Pandolf/Santee and Looney equations ($P = 0.005$ and $P = 0.009$, respectively). The totals predicted by all four of the equations differed from one another statistically with the lone exception of the ACSM and Pandolf/Santee equations totals not being statistically different ($P > 0.99$).

Figure 6.

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The mean total trial oxygen uptake predicted by each of four laboratory prediction equations based on walking speed and grade inputs measured from the trials completed by each subject ($n=7$, four left-most bars) in the unloaded condition vs. the mean values measured during these trials (dashed lines). Mean total trial oxygen uptake predicted by the two laboratory prediction equations with load capabilities ($n=6$, two right-most bars) vs. the mean measured rates of oxygen uptake. The darker dashed line represents the mean values after correction for the portable vs. laboratory calorimeter offset presented in Fig. 3; the lighter dashed line represents the mean original values measured by the portable calorimeter. [Note, the dashed line appearing within the ACSM prediction bar represents the total prediction from this laboratory equation when the downhill grades are adjusted downward and not treated as level values per the original equation (see text)]. *Significance at the 0.05 level vs. measured values.

Group mean total oxygen uptake values ($\text{mL O}_2/\text{kg}$) obtained from the field trials for the six subjects completing the loaded trials are also shown in Fig. 6 (rightmost two bars, dashed black line). The mean total obtained from the field for the loaded condition of $1,310 \pm 153 \text{ mL O}_2/\text{kg}$ was 34% greater than that from the unloaded trial measured mean. The totals predicted by the Minimum Mechanics equation of $1,289 \pm 22 \text{ mL O}_2/\text{kg}$ agreed closely, being -2% below the measured total whereas the Pandolf/Santee predicted mean of $1,463 \pm 32 \text{ mL O}_2/\text{kg}$ was 12% greater than the measured value. The Minimum Mechanics and Pandolf/Santee oxygen uptake totals did not differ statistically from the measured values ($P > 0.99$ and $P = 0.115$, respectively), but did differ from one another ($P < 0.001$; ANOVA $F = 66.7$, $df = 2$).

Standard errors of estimate (SEE) for each equation for the unloaded trials were: 107, 207, 186, and 248 $\text{mL O}_2/\text{kg}$ for the Minimum Mechanics, Pandolf/Santee, ACSM, and Looney equations, respectively; loaded trials SEE values were 176 and 241 $\text{mL O}_2/\text{kg}$ for the Minimum Mechanics and Pandolf/Santee equations, respectively.

Equation Predictions—Examples from Single Unloaded and Loaded Trials

The predicted rates of oxygen uptake per kg body weight throughout the representative field trials for a single subject in both the loaded and

unloaded conditions are shown in Fig. 7, *A* and *B*, respectively. The rates predicted throughout these trials largely parallel the differences obtained for the groups mean totals from the different equations. For example, for the unloaded trial in Fig. 7A, the rates of oxygen uptake predicted by the Minimum Mechanics model (black line) are the lowest overall while those of the Looney et al. equation are the greatest overall (blue line). The ACSM (red line) and Pandolf/Santee (green) equations predict oxygen uptake rates that are in between Minimum Mechanics and Looney equations. The Pandolf/Santee equation, which includes a downhill capability, predicts greater rates of oxygen uptake on the level course portions versus ACSM, but lower rates on the declined portions of the course. The latter observation is attributable to the ACSM equation predicting similar rates of oxygen uptake for the level and declined portions of the field course due to the absence of a declined term and consequent need to treat downhill gradients as level surfaces for predictive purposes.

Figure 7.

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Oxygen uptake rates predicted by the four laboratory equations from same GPS acquired inputs for walking speed for an individual unloaded walking field trial (A) and by the two laboratory prediction equations that include load for an individual loaded field trial (B). The Pandolf, ACSM, and Minimum Mechanics laboratory equations all predict similar values for the level and uphill portions of the course but differ on the downhill portions. The equation of Looney and colleagues predicted greater values than the other three laboratory equations over the majority of the course, and particularly on the level and downhill segments of the trial.

Between-equation prediction outputs for the Minimum Mechanics and Pandolf/Santee equations capable of loaded predictions illustrate similar patterns in Fig. 7B. Throughout this trial, the oxygen uptake rates predicted by the Minimum Mechanics equation are slightly to moderately lower than those of the Pandolf/Santee equation. The largest between-equation differences are present on the downhill portions of the course.

Constant Speed Hypothetical Trial Prediction Outcomes: Between-Equation Effects

The total oxygen uptake each of the four laboratory equations predicted to complete our simulated 6,400-m, zero net elevation change courses of shallow, moderate, and relatively steep gradients at constant speeds ranging from 0.6 to 1.8 m/s are shown in Fig. 8. For the two shallower grade courses, there is a general tendency for the values predicted by the ACSM, Minimum Mechanics, and Pandolf/Santee equations to be similar in the intermediate walking speeds from 1.1 to 1.4 m/s, whereas the Looney et al. equation predicts appreciably greater totals. On the relatively steep grade course, the Minimum Mechanics and Pandolf/Santee equations predictions are similar at lower values whereas the ACSM and Looney et al. equations have somewhat greater values that are similar to one another.

Figure 8.

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Oxygen uptake totals (mL O₂/kg) predicted by the four laboratory equations for hypothetical, constant-speed walking trials covering the same distance of just under 6,500 m at different constant speeds. Predictions were generated for hypothetical, constant-speed trials on each of three different course gradient profiles (Shallow, *A*; Moderate, *B*; Steeper, *C*). Each of the theoretical courses included level, up and downhill gradients of segment lengths equal to those illustrated in Fig. 1 for our experimental course. The three courses thus differed only in the steepness implemented for the up and downhill gradients of the constant-length graded segments. The data point in (*B*) represents the oxygen uptake mean measured for the unloaded field trials on this course. The shaded region on this panel spans the range of walking speeds individual subjects averaged over the course of their individual trials.

Each of the prediction equations exhibits different across-speed variation in the total oxygen uptake values predicted regardless of the gradient of the course. The totals predicted by the ACSM equation are greatest at slow speeds and least at the fastest speeds in the range. The totals predicted by the Minimum Mechanics equation exhibit a somewhat similar tendency to decrease with speed but do so to a lesser extent, with decreases of roughly 15% from the slowest to the fastest speed in the range examined. The Pandolf/Santee equation predicts totals that vary little across the slow and intermediate speeds but become slightly greater at the faster speeds examined. The Looney et al. equation prediction totals varied the most across speed, exhibiting differences of up to 37% across the analysis range of speeds. For the large majority of speeds and grades considered, the Looney et al. equation predicted totals that were greater than those of the other three equations.

On all three courses, the Pandolf/Santee equation predicted totals lower than the Minimum Mechanics equation at speeds below 1.0 m/s and greater totals at speeds of 1.2 m/s and above.

DISCUSSION

Our expectation that laboratory equations could accurately predict the energy expended during free walking in the field across variable gradients and terrain was well-supported, but less fully than expected. We found that that only one of four predictive equations tested met our hypothesized expectation well. Under unloaded and loaded field trial conditions, the Minimum Mechanics equation predicted the energy expended in the field to within +4% and –2% of the respective measured means (Fig. 6). By comparison, the minimum error observed for field predictions by any of the other three equations was +12%. The finding that one equation agreed to within an average of 3% across load conditions with appreciably different energetic requirements leads us to conclude that free-walking field energy expenditure can indeed be accurately predicted from a laboratory treadmill-based equation.

The unexpected outcome was the extent to which the energy expenditure predictions of the four laboratory equations tested were not similar. In contrast to the null outcome hypothesized, predictions for both load conditions differed appreciably across equation (Fig. 6). This was the case even though the speed, grade, and body weight inputs into the four equations were identical, and as noted, all four predictive equations were formulated using indirect calorimetry data obtained during constant-speed laboratory walking trials. The predictive error for the unloaded trials spanned a 16% range from a low of +4% for the Minimum Mechanics equation to a high of +20% for the Looney et al. equation, with the ACSM and Pandolf/Santee equations overpredicting by +13% and +17%, respectively. Between-equation differences for our computational comparisons that spanned a much broader range of speed and gradient conditions were, in some cases, approximately twice as large, reaching a maximum of 37%. Across the slower portion of the speed range examined from 0.6 to 1.0 m/s,

between-equation differences between the Pandolf/Santee and Looney equations consistently exceeded 20% and averaged roughly 30% regardless of the gradient profile of the course (Fig. 8). Clearly, differences in the bases of the predictive equations tested, whether formulaic or due to dataset bias, or both, can result in substantial differences in the walking energetic costs predicted from identical inputs.

Validity of the Field Energy Expenditure Data

Validity limitations for field measurements of walking energy expenditure, particularly for longer duration trials, have long been a significant challenge. However, the well-controlled validation studies on the portable calorimeter we used suggested accurate determination of field walking energy expenditure should be possible. The Douglas bag, mass spectrometer, and other laboratory-based metabolic system assessments of the validity of the Cosmed K4b₂ unit (22–28) we used reported positive, but consistent within-study bias ranging from 2% to 25% across the relevant measurement range. Compared with our Parvomedics TrueOne laboratory system quantified a consistent positive bias in the middle of the literature-reported range for our Cosmed K4b₂ unit during walking (Fig. 3) of $+13.8 \pm 1.8\%$ (means $\pm$ SD, range: +11.4% to +16.5%). The between-calorimeter offset observed for the standing trials we administered was similar at +14.3%. In addition, the level walking values acquired from the portable calorimeter at the outset of the field trials also exhibited essentially the same positive offset versus the laboratory system treadmill values (open triangles, Fig. 3). This last result would be expected from the consistency of the portable calorimeter offset as well as the absence of pavement versus treadmill walking economy differences at these speeds (Fig. 2). Thus, our collective experimental results suggest that the maximum error present on our postcorrection walking energy expenditure field means was likely $\pm 2.6\%$ or less.

Of course, we recognized that the correction factor implemented for the portable calorimeter could only be as valid as the laboratory calorimeter used to quantify it. Numerous observations support the validity of our Parvomedics TrueOne laboratory system for doing so. First, our direct comparison of our laboratory system against the gold standard Douglas bag technique during same-condition treadmill walking trials produced virtually identical values at each of three walking speeds (Fig. 5). Second, our prior validation of our laboratory system with precision infusions of N₂-CO₂ gas indicated an oxygen uptake measurement accuracy of $\pm 2.9\%$ within the relevant measurement range for walking (33). Third, the rigorous validation of the Parvomedics TrueOne system by several groups of investigators indicated the accuracy of this system all but equals that of the gold standard Douglas bag method (34, 35) in keeping with our original data here.

One important analytic note regarding our results is that we only analyzed equation accuracy of the group mean data acquired. In this regard, we deviated from the general practice (14, 15) of quantifying the accuracy of prediction for individual subjects. We did not do so here because of the small sample sizes necessitated by our demanding field-testing protocols. Equally important, accuracy metrics for individuals could misrepresent equation predictive accuracy because the measurement variability of the portable calorimeter (22–28, Fig. 3) would be present to an unknown extent.

What Explains Between-Equation Differences in Predicted Energy Expenditure?

We next consider the relative field efficacy of the four equations evaluated here in order of their appearance in the literature. We do so with a primary reliance on our field walking results, the treadmill/terrain factors quantified here, and prior treadmill results (14) that assessed the accuracy of three of the four equations analyzed here across 40 field-relevant speed, grade, and

load conditions.

The ACSM Equation

The ACSM equation is the briefest and simplest of the four equations and was second most accurate in predicting the energy expended for the unloaded field trials. The conciseness of the equation is possible because neither load nor downhill gradients are included and the $\dot{V}O_2$ versus speed relationship is described as a linear function on all gradients. The linear treatment of a relationship well-known to be curvilinear (14, 36–38) does compromise predictive accuracy somewhat across speed, but to a relatively small degree (36, Fig. 4C).

The ACSM equation field overestimation of +13% is clearly at least somewhat attributable to the absence of negative gradient, or downhill, capabilities. In the present study, this limitation required us to treat the downhill sections of the field course as level for predictive purposes. Doing so undoubtedly resulted in predicted rates of energy expenditure that were generally greater than those of the other three equations for the downhill portions of the course (Fig. 7A) thereby contributing to the over-prediction. We eliminated this limitation by implementing the simple downhill coefficient validated in an earlier study (14) of 0.73 times the walking portion of the body's metabolic rate (i.e., total $\dot{V}O_2$ minus resting $\dot{V}O_2$) on level gradients. This lone adjustment reduced the overprediction of the ACSM equation by 5%. The remaining +8% predictive error seems most likely attributable to consistent overpredictions observed across all positive grades under treadmill conditions (14, Fig. 4C).

One notable benefit of the conciseness of the ACSM equation is that its simplicity allows for straightforward modifications for downhill gradients as noted earlier (coefficient=0.73) and for load carriage also (load mass = body

mass) as implemented elsewhere (14). When both modifications were implemented to generate predictions for the loaded field trials here, the predicted total exceeded the measured field mean by only +8%.

The Pandolf/Santee Equation

In contrast to the ACSM equation, the relatively lengthy Pandolf/Santee equation resulted from its purposeful evolution to include the speed, grade, load and terrain capabilities necessary for field predictions. The multistep evolution that included Goldman and Lampietro (7), Givoni and Goldman (6), and Soule et al. (8) before the original Pandolf et al. equation paper (5) and subsequent extension by Santee et al. (9) to include negative gradients, ultimately resulted in a lengthy equation requiring multiple inputs of the speed, grade, and load to encompass specific trial conditions. Nonetheless, and despite a somewhat limited empirical basis for negative gradients, the full equation performs relatively accurately under laboratory conditions, particularly across speed on the declines tested (14, Fig. 4E and Fig. 5E).

The respective overestimates of +17% and +12% for the unloaded and loaded field trials of the Pandolf/Santee equation in the field here resulted predominantly from moderate overpredictions on all portions of the field course (negative, level, and positive grades) in the range of relatively fast walking speeds subjects chose. Predicted rates of energy expenditure exceeding those of the ACSM and Minimum Mechanics equations were present throughout but were most pronounced for the inclined portions of the course (Fig. 7) as our laboratory tests suggested would be the case (14, Fig. 4C). In keeping with our prior laboratory testing results, the Pandolf/Santee equation provided slightly more accurate predictions under loaded versus unloaded conditions in the field. This was primarily because the 30% body weight loads administered in our field tests here increased walking rates of energy expenditure slightly more than 30%, a phenomenon

predicted by the nonlinear treatment of load in the Pandolf/Santee equation.

The Minimum Mechanics Equation

The Minimum Mechanics equation is moderate in length and complexity and is the only one of the four equations that was formulated from a single dataset incorporating speed, grade, and load and also acquired for the purpose of developing a predictive equation. Numerous factors appear to have contributed to the accuracy of the unloaded and loaded field means predicted here using this equation. These include dedicated equation development using a broad, field-condition relevant dataset acquired with a well-validated metabolic system, the incorporation of directly measured terrain factors, careful correction of the measurement bias of the portable calorimeter used in the field, and the predictive accuracy previously developed and validated under laboratory conditions.

Compared with each of the other three equations, the Minimum Mechanics equation predicted rates of walking energy expenditure that were somewhat lower across all portions of the course (Fig. 7). The treatment of each kg of load carried as body weight resulted in the Minimum Mechanics equation predictions that were largely accurate, being relatively lower for the loaded versus unloaded condition due to load effects slightly greater than the 30% predicted. The use of a single coefficient of 0.73 times level values for downhill walking also appeared to be relatively accurate on the -3% gradients on our course.

The Looney Equation

The Looney et al. equation has two features that distinguish it from the other three equations examined here. First, unlike the Minimum Mechanics and Pandolf/Santee equations, the Looney et al. equation does not adopt a binary

approach to positive and negative gradients (Table 1). Rather, positive and negative gradients are described in a single equation by using multiple quadratic terms, making it the most complex of those considered here. Second, this equation is the only one of the four that was formulated with data that were not originally acquired for the purpose of developing a predictive equation. Rather, Looney et al. aggregated data from the literature across numerous studies of differing experimental purposes. Doing so enabled these investigators to span a broader range of gradients, but also eliminated the ability to control measurement error that direct experimentation provides. Although the authors refer to the Looney equation as a load carriage tool, the published equation does not incorporate load. At present, how a component or modifier to include load might be introduced into the existing equation is unclear. Accordingly, we only examined the performance of the Looney et al. equation for the unloaded portion of our study.

The combination of reliance on literature data and a single mathematical description for both positive and negative grades results in the Looney equation having consistent positive bias (Figs. 7A and 8B) that results in the +20% overprediction observed for the unloaded trials. The comparisons available indicate a moderate positive bias during level and inclined portions of the field course and a somewhat greater positive bias on the declined portions. On the -3% gradient segments of our field course, the rates of energy expenditure predicted by the Looney equation generally equaled those predicted for level conditions by the ACSM equation, and substantially exceeded those predicted by both Minimum Mechanics and Pandolf/Santee equations (Fig. 7).

Additional Practical Equation Considerations

Beyond the respective simplicity and accuracy advantages identified for the

ACSM and Minimum Mechanics equations, several less obvious but consequential features should be noted. First, load carriage results at and above the 30% body weight loads included here (8, 14, 32, 39) raise the possibility that the Pandolf/Santee equation may become relatively more accurate under more heavily loaded conditions, (i.e., >30% body weight). Second, we cannot rule out the possibility that the Looney equation, despite a high overall bias, may perform relatively better during unloaded walking on very steep declines. Third, at walking speeds of less than 0.6 m/s, the Pandolf/Santee equation is likely the most accurate of the four evaluated. Although the Pandolf/Santee equation is likely biased slightly low throughout this very slow speed range (14, Figs. 4 and 5E) even when the lower cost of asphalt versus treadmill walking at slower speeds measured here is considered (Fig. 2; 18), the equation formulation avoids the progressive overpredictions of each of the other three equations as walking speeds approach zero (Fig. 8).

Computational Comparisons

Our computational efforts substantially expanded the range of speeds and gradients for between-equation comparisons (Fig. 8), but also lacked an empirical reference for accuracy (Fig. 8B), excepting those values acquired at the speeds subjects selected in the field for the unloaded (shaded speed range and mean, Fig. 8B) and loaded field load trials. The comparisons available across the speed range examined reflect the formulaic differences noted for the respective equations provided in Table 1. The linear treatment of the $\dot{V}O_2$ -speed relationship added to a constant resting baseline value in the case of the ACSM equation results in near-linear decreases in total energy required to cover the 4-mile course at faster speeds. In contrast, the relatively steep slope of the Pandolf/Santee equation overpredicts the increase in the energy expenditure totals over the faster two-thirds of this

speed range.

More importantly, these comparisons reveal that the energy expenditure totals predicted can vary by as much as 37%, with discrepancies that were generally most pronounced over the slower portion of the speed range examined. Accordingly, the equation chosen for field applications, particularly for hikes and marches at slower speeds are more than large enough to be of practical consequence. Based on the empirical information available, the Pandolf/Santee equation (14, Fig. 4C) is probably moderately low at the slower speeds, but less so than measured at these speeds in the laboratory due to the relatively lower rates of energy expenditure we observed for overground versus treadmill walking at slower speeds here (Fig. 2). The Looney and ACSM equations both appear to have appreciable positive bias at and below 1.0 m/s. At speeds above 1.4 m/s, both Pandolf/Santee and Looney equations appear to have positive bias. This results from a relatively steep slope for the $\dot{V}O_2$ -speed relation in the case of the Pandolf/Santee equation, and the overall positive bias for the Looney equation regardless of speed.

Concluding Remarks

In closing, we offer a first conclusion that is partially positive and a second that is largely cautionary. First, accurate prediction of walking energy expenditure in varied terrain field settings using a laboratory-derived prediction equation clearly is possible as prediction accuracy under two load conditions was $\leq 4\%$ for one of the four equations tested. Second, the choice of existing equations, due to derivation and formulation constraints, can have a large impact on the validity of the predictions generated, even when using identical inputs. Thus, the accuracy obtained when using laboratory-based equations in the field is significantly equation-dependent.

For practicality, we have reduced the present results to a concise reference for hikers, soldiers, wildland firefighters, and others who might seek out guidance on field walking energy expenditure estimation (Table 3). Our tabular reference for equation efficacy includes three factors: empirical accuracy, breadth of field application, and ease of use.

Table 3.

Literature predictive equations rank ordered for predictive accuracy, conditional breadth in the field, and ease of use

Equation	Accuracy	Field Breadth	Ease of Use	Average Rank
ACSM	2	4	1	2.3
Looney	4	3	3	3.3
Minimum Mechanics	1	1	2	1.3
Pandolf/Santee	2	1	3	2.0

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Breadth of field application was based on the incorporation of four factors: speed, grade, negative gradients, and load conditions or lack thereof. Ease of use was based on the number of terms in the predictive equations. Predictive accuracy was based on the measured vs. predicted agreement reported in the results.

Finally, on a more basic level our results support the long-standing assumption that the energy cost of free walking in the field can be accurately inferred from constant-speed, laboratory-treadmill measurements. Specifically, the robust empirical support under free walking field conditions here corroborates the long-standing, multidisciplinary view (1, 3) that adaptative pressures economizing walking importantly influenced human physical evolution.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

P.G.W. and L.W.L. conceived and designed research; L.W.L., J.J.N., and M.J.B. performed experiments; P.G.W., L.W.L., and M.J.B. analyzed data; P.G.W., J.J.N., and M.J.B. interpreted results of experiments; P.G.W. and M.J.B. prepared figures; P.G.W. drafted manuscript; P.G.W., L.W.L., and M.J.B. edited and revised manuscript; P.G.W., L.W.L., and M.J.B. approved final version of manuscript.

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REFERENCES

- 1.Alexander RM. *Energetics and the optimization of human walking and running: the Raymond Pearl memorial lecture*. *Am J Hum Biol* 14: 641–648, 2002. doi: 10.1002/ajhb.10067. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
- 2.Rodman PS, McHenry HM. *Bioenergetics and the origin of hominid bipedalism*. *Am J Phys Anthropol*, 52: 103–106, 1980. doi:

- 10.1002/ajpa.1330520113. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
- 3.Sockol MD, Raichlen DA, Pontzer H. Chimpanzee locomotor energetics and the origin of human bipedalism. *Proc Natl Acad Sci USA* 104: 12265–12269, 2007. doi: 10.1073/pnas.0703267104. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
- 4.Sol JA, Ruby BC, Gaskill SE, Dumke CL, Domitrovich JW. Metabolic demand of hiking in wildland firefighting. *Wilderness Environl Med* 29: 304–314, 2018. doi: 10.1016/j.wem.2018.03.006. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
- 5.Pandolf KB, Givoni B, Goldman RF. Predicting energy expenditure with loads while standing or walking very slowly. *J Appl Physiol Respir Environ Exerc Physiol* 43: 577–581, 1977. doi: 10.1152/jappl.1977.43.4.577. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
- 6.Givoni B, Goldman RF. Predicting metabolic energy costs. *J Appl Physiol* 30: 429–433, 1971. doi: 10.1152/jappl.1971.30.3.429. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
- 7.Goldman RF, Iampietro PF. Energy cost of load carriage. *J Appl Physiol* 17: 675–676, 1962. doi: 10.1152/jappl.1962.17.4.675. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
- 8.Soule RG, Pandolf KB, Goldman RF. Energy expenditure of heavy load carriage. *Ergonomics* 21: 373–381, 1978. doi: 10.1080/00140137808931734. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
- 9.Santee WR, Blanchard LA, Speckman KL, Gonzalez JA, Wallace RF. Load Carriage Model Development and Testing with Field Data. Technical Note TN03-3. Natick, MA: US Army Research Institute of Environmental Medicine, 2003. [[Google Scholar](#)]
- 10.American College of Sports Medicine. *Guidelines for Graded Exercise Testing and Prescription* (7th ed.). Philadelphia, PA: Lippincott, Williams and Wilkins, 2013. [[Google Scholar](#)]
- 11.Dill DB. Oxygen used in horizontal and grade walking and running on the

- treadmill. *J Appl Physiol* 20: 19–22, 1965. doi: 10.1152/jappl.1965.20.1.19. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
- 12.Nagle F, Balke B, Naughton J. Gradational test for assessing work capacity. *J Appl Physiol* 20: 745–748, 1965. doi: 10.1152/jappl.1965.20.4.745. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
- 13.Looney DP, Santee WR, Hansen EO, Bonventre PJ, Chalmers CR, Potter AW. Estimating energy expenditure during level, uphill and downhill walking. *Med Sci Sports Exerc* 51: 1954–1969, 2019. doi: 10.1249/MSS.0000000000002002. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
- 14.Ludlow LW, Weyand PG. Walking economy is predictably determined by speed, grade and gravitational load. *J Appl Physiol* 123: 1238–1302, 2017. doi: 10.1152/japplphysiol.00504.2017. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
- 15.Looney DP, Potter AW, Pryor JL, Bremner PE, Chalmers CR, Mcclung HL, Welles AP, Santee WR. Metabolic costs of standing and walking in healthy military-age adults: a meta-regression. *Med Sci Sports Exerc* 51: 346–351, 2019. doi: 10.1249/MSS.0000000000001779. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
- 16.Lejeune TM, Willems PA, Heglund NC. Mechanics and energetics of human locomotion on sand. *J Exp Biol* 201: 2071–2080, 1998. doi: 10.1242/jeb.201.13.2071. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
- 17.Daniels F, Vanderbie JH, Winsmann FR. Energy Cost of Treadmill Walking Compared to Road Walking. Report No. 220. Natick, MA: Quartermaster Research and Engineering Command, 1953. [\[Google Scholar\]](#)
- 18.Soule RG, Goldman RF. Terrain coefficients for energy cost prediction. *J Appl Physiol* 32: 706–708, 1972. doi: 10.1152/jappl.1972.32.5.706. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
- 19.de Mullenheim PY, Dumond R, Gernigon M, Mahé G, Lavenue A, Bickert S, Prioux J, Noury-Desvaux B, Le Faucheur A. Low cost GPS prediction of walking energy expenditure in the field. *J Appl Physiol* (1985) 121: 577–588, 2016. doi: 10.1152/japplphysiol.00224.2016. [\[DOI\]](#) [\[PubMed\]](#) [\[Google\]](#)

[Scholar\]](#)

20.Looney DP, Santee WR, Blanchard LA, Rome MN, Carter AJ, Potter AW, Mullen S. Metabolic costs of military load carriage over complex terrain. *Mil Med* 183: e357–e2362, 2018. doi: 10.1093/milmed/usx099. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

21.Potter AW, Santee WR, Mullen SP, Karis AJ, Blanchard LA, Rome MN, Pitts KP, Looney DP. Complex terrain load carriage energy expenditure estimation using global positioning system devices. *Med Sci Sports Exerc* 50: 2145–2149, 2018. doi: 10.1249/MSS.0000000000001669. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

22.Duffield R, Dawson B, Pinnington HC, Wong P. Accuracy and reliability of a Cosmed K4b₂ portable gas analysis system. *J Sci Med Sport* 7: 11–22, 2004. doi: 10.1016/S1440-2440(04)80039-2. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

23.Eisenmann JC, Brisko N, Shadrack D, Welsh S. Comparative analysis of the Cosmed Quark b₂ and K4b₂ gas analysis systems during submaximal exercise. *J Sports Med Phys Fitness* 43: 150, 2003. [[PubMed](#)] [[Google Scholar](#)]

24.Kesler RM, Hsiao-Wecksler ET, Motl RW, Klaren RE, Ensari I, Horn GP. A modified SCBA mouthpiece for accurate metabolic data collection from firefighters. *Ergonomics* 58: 148–159, 2015. doi: 10.1080/00140139.2014.964783. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

25.Littlewood RA, White MS, Bell KL, Davies PSW, Cleghorn GJ, Grote R. Comparison of the Cosmed K4b₂ and Deltrac II metabolic cart in measuring resting energy expenditure in adults. *Clin Nutr* 21: 491–497, 2002. doi: 10.1054/clnu.2002.0580. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

26.McLaughlin JE, King GA, Howley ET, Bassett DR, Ainsworth BE. Validation of the Cosmed K4b portable metabolic system. *Int J Sports Med* 22: 280–284, 2001. doi: 10.1055/s-2001-13816. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

27.McNaughton LR, Sherman Roberts S, Bentley DJ. Portable gas analyser

Cosmed K4b2 compared to a laboratory based mass spectrometer system. *J Sports Med Phys Fitness* 45: 315, 2005. [[PubMed](#)] [[Google Scholar](#)]

28. Pinnington HC, Wong P, Tay J, Green D, Dawson B. The level of accuracy and agreement in measures of FEO₂, FECO₂ and VE between the Cosmed K4b2 portable, respiratory gas analysis system and a metabolic cart. *J Sci Med Sport* 4: 324–335, 2001. doi: 10.1016/S1440-2440(01)80041-4. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

29. Linnarsson D, Karlsson J, Fagraeus LE, Saltin BE. Muscle metabolites and oxygen deficit with exercise in hypoxia and hyperoxia. *J Appl Physiol* 36: 399–402, 1974. doi: 10.1152/jappl.1974.36.4.399. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

30. Whipp BJ, Seard C, Wasserman K. Oxygen deficit-oxygen debt relationships and efficiency of anaerobic work. *J Appl Physiol* 28: 452–456, 1970. doi: 10.1152/jappl.1970.28.4.452. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

31. Blaxter K. *Energy Metabolism in Animals and Man*. Cambridge: Cambridge University Press, 1989. [[Google Scholar](#)]

32. Vine CA, Coakley SL, Blacker SD, Doherty J, Hale BJ, Walker EF, Rue CA, Lee BJ, Flood TR, Knapik JJ, Jackson S. Accuracy of metabolic cost predictive equations during military load carriage. *J Strength Cond Res*. In press. doi: 10.1519/JSC.0000000000003644. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

33. Weyand PG, Smith BR, Puyau MR, Butte NF. The mass-specific energy cost of human walking is set by stature. *J Exp Biol* 213: 3972–3979, 2010. doi: 10.1242/jeb.048199. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

34. Bassett DR, Howley ET, Thompson DL, King GA, Strath SJ, McLaughlin JE, Parr BB. Validity of inspiratory and expiratory methods of measuring gas exchange with a computerized system. *J Appl Physiol* (1985) 91: 218–224, 2001. doi: 10.1152/jappl.2001.91.1.218. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

35. Crouter SE, Antczak A, Hudak JR, DellaValle DM, Haas JD. Accuracy and reliability of the ParvoMedics TrueOne 2400 and MedGraphics VO2000

- metabolic systems. Eur J Appl Physiol* 98: 139–151, 2006. doi: 10.1007/s00421-006-0255-0. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
- 36.Ludlow LW, Weyand PG. Estimating loaded, inclined walking energetics: no functional difference between added and body mass. *IEEE 13th International Conference on Wearable and Implantable Body Sensor Networks (BSN)*. 306–311, 2016. doi: 10.1109/BSN.2016.7516279. [[DOI](#)]
- 37.Margaria R. Sulla fisiologica e specialmente sul consumo energetico della corsa a varie velocita ed inclinazioni. *Atti Accademia Nazionale dei Lincei* 7: 299–368, 1938. [[Google Scholar](#)]
- 38.Margaria R, Cerretelli P, Aghemo P, Sassi G. Energy cost of running. *J Appl Physiol* 18: 367–370, 1963. doi: 10.1152/jappl.1963.18.2.367. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
- 39.Drain JR, Aisbett B, Lewis M, Billing DC. The Pandolf equation underpredicts the metabolic rate of contemporary military load carriage. *J Sci Med Sport* 20: S104–S108, 2017. doi: 10.1016/j.jsams.2017.08.009. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

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